Chapter 7. Finite Fourier Analysis

Fourier analysis on Z(N) and finite abelian groups

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7.1 Fourier analysis on $\mathbb{Z}(N)$

Motivation. Replacing the circle by the N-th roots of unity, Fourier series become finite sums: inversion and Parseval are plain linear algebra with an orthogonal basis of exponentials. Splitting into even and odd indices computes all N coefficients in $O(N \log N)$ operations (the fast Fourier transform).

7.1.1 The group $\mathbb{Z}(N)$

For $N \geq 1$ let $\zeta = e^{2\pi i/N}$. The N-th roots of unity $\{1, \zeta, \ldots, \zeta^{N-1}\}$ form a group under multiplication, and $\zeta^k \mapsto k$ identifies it with $\mathbb{Z}(N) = \mathbb{Z}/N\mathbb{Z}$ under addition mod N. Functions on $\mathbb{Z}(N)$ are N-periodic functions on $\mathbb{Z}$; one can view $\mathbb{Z}(N)$ as N equally spaced points on the circle (*discretization*).

7.1.2 Fourier inversion theorem and Plancherel identity on $\mathbb{Z}(N)$

V, the complex functions on $\mathbb{Z}(N)$, is an N-dimensional vector space with inner product $(F, G) = \sum_{k=0}^{N-1} F(k)\overline{G(k)}$ and norm $\|F\|^2 = \sum |F(k)|^2$. For $\ell = 0, \ldots, N-1$ let $e_\ell(k) = \zeta^{\ell k}$.

Lemma 1.1

The family $\{e_0, \ldots, e_{N-1}\}$ is orthogonal: $(e_m, e_\ell) = N$ if $m = \ell$ and 0 if $m \neq \ell$. Hence $\{e_\ell/\sqrt{N}\}$ is an orthonormal basis of V.

Proof. $(e_m, e_\ell) = \sum_{k=0}^{N-1} \zeta^{(m-\ell)k}$. If $m = \ell$ this is N. Otherwise $q = \zeta^{m-\ell} \neq 1$ and $q^N = 1$, so the geometric sum is $\frac{1-q^N}{1-q} = 0$. N orthogonal nonzero vectors in an N-dimensional space form a basis. ■

The n-th **Fourier coefficient** of F is $a_n = \frac{1}{N}(F, e_n) = \frac{1}{N}\sum_{k=0}^{N-1} F(k)\zeta^{-kn}$.

Theorem 1.2

If F is a function on $\mathbb{Z}(N)$, then $F(k) = \sum_{n=0}^{N-1} a_n \zeta^{nk}$ (inversion) and $\sum_{n=0}^{N-1} |a_n|^2 = \frac{1}{N}\sum_{k=0}^{N-1} |F(k)|^2$ (Parseval).

Proof. Expand F in the orthonormal basis $e_n^* = e_n/\sqrt{N}$: $F = \sum (F, e_n^*)e_n^* = \sum \frac{1}{N}(F, e_n)e_n = \sum a_n e_n$. Pythagoras gives $\|F\|^2 = \sum |(F, e_n^*)|^2 = \sum N|a_n|^2$. ■

(Discrete analogue of Fourier series: if $F(k) = f(k/N)$ for a 1-periodic f, then a_n is a Riemann sum for $\hat{f}(n)$.)

7.1.3 The fast Fourier transform

Computing all a_n naively takes about N^2 operations. Let $\#(M)$ be the minimal number of operations (additions and multiplications) needed to compute all Fourier coefficients of any function on $\mathbb{Z}(M)$, here with $\omega_M = e^{-2\pi i/M}$, so that $a_k^M(F) = \frac{1}{M}\sum_r F(r)\omega_M^{kr}$.

Theorem 1.3

For $N = 2^n$, the Fourier coefficients of a function on $\mathbb{Z}(N)$ can be computed with at most $4 \cdot 2^n n = 4N \log_2 N = O(N \log N)$ operations, given ω_N.

Lemma 1.4

Given ω_{2M}, $\#(2M) \le 2\#(M) + 8M$.

Proof. Computing $\omega_{2M}, \omega_{2M}^2, \dots, \omega_{2M}^{2M}$ takes $\le 2M$ operations, and gives $\omega_M = \omega_{2M}^2$. For F on $\mathbb{Z}(2M)$ put $F_0(r) = F(2r)$, $F_1(r) = F(2r+1)$, functions on $\mathbb{Z}(M)$. Splitting the sum into even and odd indices,

$$a_k^{2M}(F) = \frac{1}{2M}\sum_{r=0}^{2M-1} F(r)\omega_{2M}^{kr} = \frac{1}{2}\Big(\frac{1}{M}\sum_{\ell} F_0(\ell)\omega_M^{k\ell} + \frac{1}{M}\sum_{m} F_1(m)\omega_M^{km}\omega_{2M}^{k}\Big)$$
$$= \frac{1}{2}\big(a_k^M(F_0) + a_k^M(F_1)\omega_{2M}^k\big)$$

(with a_k^M read M-periodically in k). Computing all $a^M(F_0)$, $a^M(F_1)$ costs $2\#(M)$; then each of the $2M$ coefficients costs 3 operations. So $\#(2M) \le 2M + 2\#(M) + 6M$. ∎

Proof of Theorem 1.3. Induction on n. For $N = 2$: $a_0 = \frac{1}{2}(F(0)+F(1))$, $a_1 = \frac{1}{2}(F(0)-F(1))$, at most $5 \le 8$ operations. If $\#(2^{n-1}) \le 4\cdot 2^{n-1}(n-1)$, then by Lemma 1.4, $\#(2^n) \le 2\cdot 4\cdot 2^{n-1}(n-1) + 8\cdot 2^{n-1} = 4\cdot 2^n n$. ∎

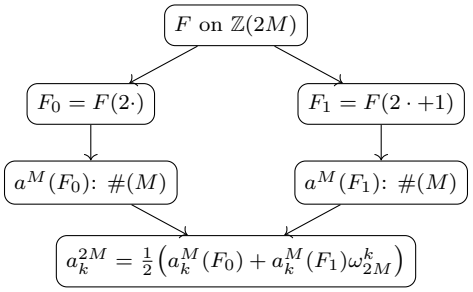


Figure. *One step of the fast Fourier transform (Lemma 1.4).*

7.2 Fourier analysis on finite abelian groups

Motivation. On a general finite abelian group the role of the exponentials is played by characters, homomorphisms into the unit circle. They are orthonormal, and there are enough of them to form a basis (simultaneous diagonalization of the translation operators). This gives Fourier inversion and Plancherel on every finite abelian group, the tool needed for Dirichlet's theorem.

7.2.1 Abelian groups

An **abelian group** $(G, \cdot)$ is associative and commutative with identity 1 and inverses. Examples: $\mathbb{Z}(N)$; the circle $\mathbb{R}/\mathbb{Z}$; $\mathbb{R}$, $\mathbb{R}^+ = (0,\infty)$ under multiplication; $\mathbb{Z}^*(q)$, the units of $\mathbb{Z}(q)$, i.e. residues coprime to q, under multiplication. A **homomorphism** $f : G \to H$ satisfies $f(ab) = f(a)f(b)$; an **isomorphism** is a bijective homomorphism. The **direct product** $G_1 \times G_2$ has componentwise multiplication. (E.g. $\mathbb{Z}^*(8) \cong \mathbb{Z}(2)\times\mathbb{Z}(2)$, while $\mathbb{Z}^*(5) \cong \mathbb{Z}(4)$.) Every finite abelian group is a direct product of cyclic groups; we will not need this.

7.2.2 Characters

A **character** of G is a function $e : G \to S^1 = \{|z| = 1\}$ with $e(ab) = e(a)e(b)$, i.e. a homomorphism $G \to S^1$. The **trivial character** is $e \equiv 1$. $\hat{G}$ denotes the set of characters. On $\mathbb{Z}(N)$ the characters are exactly $e_\ell(k) = \zeta^{\ell k}$ (a character is determined by $e(1)$, an N-th root of unity).

Lemma 2.1

$\hat{G}$ is an abelian group under $(e_1e_2)(a) = e_1(a)e_2(a)$.

Proof. Products of characters are characters; the trivial character is the identity; the inverse of e is $\bar{e} = 1/e$ (as $|e| = 1$); associativity and commutativity are pointwise. ■

Lemma 2.2

If G is finite and $e : G \to \mathbb{C} - \{0\}$ satisfies $e(ab) = e(a)e(b)$, then e is a character.

Proof. G is finite, so $|e(a)|$ is bounded above and below by positive numbers. If $|e(a)| > 1$, then $|e(a^n)| = |e(a)|^n \to \infty$, impossible; if $|e(a)| < 1$, $|e(a^n)| \to 0$, also impossible. So $|e(a)| = 1$. ■

7.2.3 The orthogonality relations

On the space V of complex functions on finite G use $(f, g) = \frac{1}{|G|}\sum_{a\in G} f(a)\overline{g(a)}$.

Theorem 2.3

The characters of G form an orthonormal family in V.

Lemma 2.4

If e is a nontrivial character, then $\sum_{a\in G} e(a) = 0$.

Proof. Choose b with $e(b) \neq 1$. Since $a \mapsto ba$ permutes G, $e(b)\sum_a e(a) = \sum_a e(ba) = \sum_a e(a)$; so $\sum_a e(a) = 0$. ■

Proof of Theorem 2.3. $(e, e) = \frac{1}{|G|}\sum |e(a)|^2 = 1$. For $e \neq e'$, $e\overline{e'} = e(e')^{-1}$ is a nontrivial character, so $(e, e') = \frac{1}{|G|}\sum_a e(a)\overline{e'(a)} = 0$ by Lemma 2.4. ■

Consequently $|\hat{G}| \leq |G|$ (orthonormal vectors are independent).

7.2.4 Characters as a total family

Theorem 2.5

The characters of a finite abelian group G form a basis of the vector space V of functions on G; in particular $|\hat{G}| = |G|$.

Lemma 2.6

A commuting family $\{T_1, \ldots, T_k\}$ of unitary transformations of a finite-dimensional inner product space V ($T_iT_j = T_jT_i$) is simultaneously diagonalizable: V has a basis of common eigenvectors.

Proof. Induction on k. For $k = 1$ this is the spectral theorem for unitary operators. Write $V = V_{\lambda_1} \oplus \cdots \oplus V_{\lambda_s}$, the eigenspaces of T_k. Each T_j, $j < k$, maps V_{λ_i} into itself: if $T_kv = \lambda_i v$, then $T_kT_jv = T_jT_kv = \lambda_iT_jv$. The restrictions of $T_1, \ldots, T_{k-1}$ to V_{λ_i} are commuting unitary operators, so by induction V_{λ_i} has a basis of common eigenvectors. Their union is the desired basis of V. ■

Proof of Theorem 2.5. For $a \in G$ define T_a on V by $(T_a f)(x) = f(ax)$. $T_a T_b = T_b T_a$ (G abelian), and T_a is unitary since $x \mapsto ax$ permutes G. By Lemma 2.6 there is a basis $\{v_b\}$ of common eigenvectors, $T_a v = \lambda_a v$. For such v: $v(1) \neq 0$, since otherwise $v(a) = (T_a v)(1) = \lambda_a v(1) = 0$ for all a. Let $w(x) = v(x)/v(1)$. Then $w(x) = \lambda_x$ (as $v(x) = (T_x v)(1) = \lambda_x v(1)$), so $w(ab) = \lambda_{ab}$, and $\lambda_{ab} v = T_{ab} v = T_a T_b v = \lambda_a \lambda_b v$ gives $w(ab) = w(a)w(b)$; also $w(x) \neq 0$ (as T_x is invertible). By Lemma 2.2, w is a character. The basis $\{v_b\}$ consists of multiples of characters, so the characters span V; with Theorem 2.3 they form a basis. ∎

7.2.5 Fourier inversion and Plancherel formula

For f on G and $e \in \hat{G}$, the **Fourier coefficient** is $\hat{f}(e) = (f, e) = \frac{1}{|G|} \sum_{a \in G} f(a)\overline{e(a)}$, and the **Fourier series** is $f \sim \sum_{e \in \hat{G}} \hat{f}(e) e$.

Theorem 2.7

The characters of a finite abelian group G form an orthonormal basis of V with $(f, g) = \frac{1}{|G|} \sum_a f(a)\overline{g(a)}$. For every f on G, $f = \sum_{e \in \hat{G}} \hat{f}(e) e$ (inversion) and $\|f\|^2 = \sum_{e \in \hat{G}} |\hat{f}(e)|^2$ (Plancherel).

Proof. Theorems 2.3 and 2.5; then expand in the orthonormal basis and apply Pythagoras. ∎

For $G = \mathbb{Z}(N)$ this recovers Theorem 1.2 (up to the normalization of the inner product).